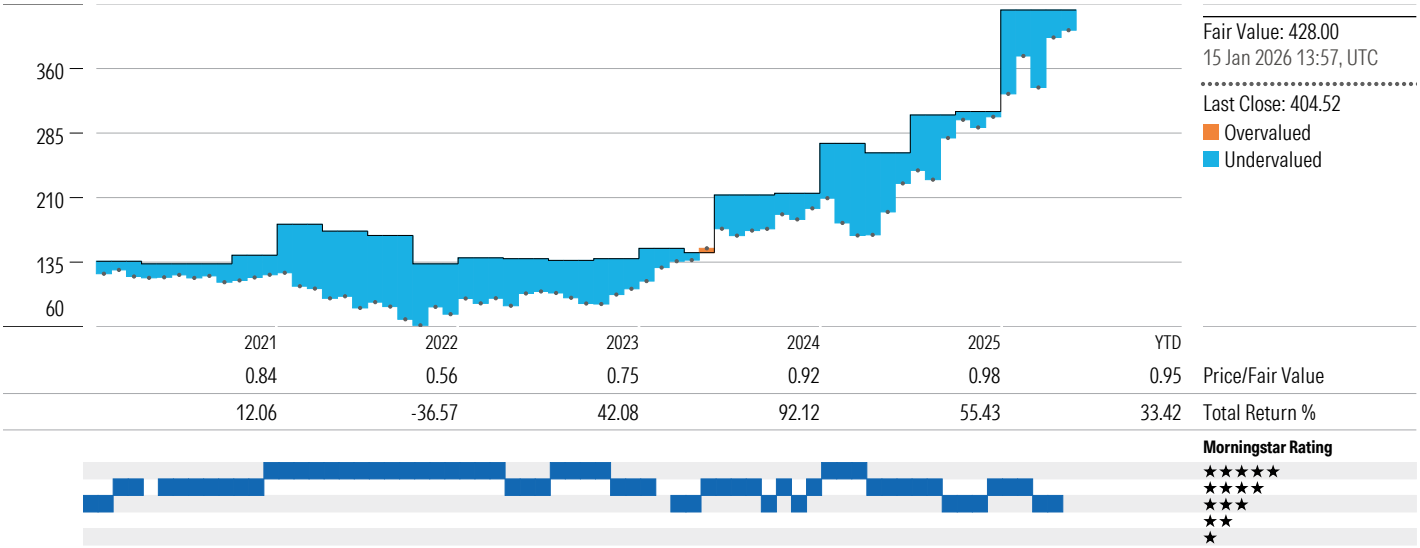
Taiwan Semiconductor Manufacturing Co Ltd ADR TSM ★★★ 22 May 2026 21:47, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
404.52 USD 22 May 2026	428.00 USD 15 Jan 2026 19:57, UTC	0.95	2.10 USD Tril 26 May 2026	Wide	Large Growth	Medium	Standard	 26 May 2026 05:00, UTC

Price vs. Fair Value



	2021	2022	2023	2024	2025	YTD	
	0.84	0.56	0.75	0.92	0.98	0.95	Price/Fair Value
	12.06	-36.57	42.08	92.12	55.43	33.42	Total Return %

Morningstar Rating

Total Return % as of 22 May 2026. Last Close as of 22 May 2026. Fair Value as of 15 Jan 2026 13:57, UTC.

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The primary analyst covering this company does not own its stock.

[1]The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

Our Foundry Outlook Is Unchanged After Huawei's Breakthrough in Improving Chip Performance

Analyst Note Phelix Lee, Senior Equity Analyst, 26 May 2026

We assess the impact on our coverage of Taiwan Semiconductor Manufacturing Company, United Microelectronics, SMIC, and Hua Hong Semiconductor after Huawei revealed its approach to improving chip performance, called Tau Scaling Law. SMIC and Hua Hong shares jumped on the news.

Why it matters: Tau scaling focuses on reducing downtime between data transmissions. It does not require extreme ultraviolet lithography tools that ASML sells. This, in theory, sidesteps export restrictions that Huawei and Chinese foundries face while potentially closing in on chip performance.

- We think Huawei's approach is not replicable, especially in the next five years. Huawei uniquely and vertically integrates chip design, chip manufacturing (via partners), consumer electronics, cars, and network infrastructure under one roof.
- The longer-term outlook skews in favor of the Chinese ecosystem spanning chipmakers, artificial intelligence application developers, and nonlithography equipment makers. But this is not at the expense of US AI spending, given current demand already overwhelms supply on many fronts.

The bottom line: We maintain our fair value estimates for TSMC, UMC, Semiconductor Manufacturing International Corporation, and Hua Hong at TWD 2,700, TWD 81, HKD 31, and HKD 38, respectively. We do not believe that Huawei's breakthrough hurts TSMC and UMC's outlook, while we already baked in bullish data center assumptions for SMIC and Hua Hong.

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Sector Technology

Industry Semiconductors

Business Description

Taiwan Semiconductor Manufacturing Co. is the world's largest dedicated chip foundry, with about 70% market share in 2025. TSMC was founded in 1987 as a joint venture of Philips, the government of Taiwan, and private investors. It went public in Taiwan in 1994 and as an ADR in the US in 1997. TSMC's scale and high-quality technology allow the firm to generate solid operating margins, even in the highly competitive foundry business. Furthermore, the shift to the fabless business model has created tailwinds for TSMC. The foundry leader has an illustrious base of customers, including Apple, AMD, and Nvidia, that look to apply its cutting-edge process technologies to their semiconductor designs. TSMC employs more than 83,000 people.

- We are skeptical that Tau scaling will be on par with a conventional 1.4-nanometer chip by 2031, as Huawei claimed. Even with design and circuitry optimizations, transistor-level performance remains constrained by 7 nm process characteristics, capping the energy efficiency of Tau scaling.
- Taken together, this validates Nvidia CEO Jensen Huang's view that US export controls are forcing China to accelerate domestic semiconductor development.

Business Strategy & Outlook Phelix Lee, Senior Equity Analyst, 15 Jan 2026

Taiwan Semiconductor Manufacturing Co. is the world's largest dedicated contract chip manufacturer, or foundry, with about 70% market share in 2025. It makes integrated circuits for customers based on their proprietary IC designs. TSMC has long benefited from semiconductor firms around the globe transitioning from integrated device manufacturers to fabless designers. Like all foundries, it assumes the costs and capital expenditures of running factories amid a highly cyclical market for its customers. Foundries tend to add excessive capacity during times of burgeoning demand, which can result in underutilization during downturns that hampers profitability.

The rise of fabless semiconductor firms has supported the growth of foundries, which has in turn encouraged increased competition. However, most of these newer competitors are confined to low-end manufacturing due to prohibitive costs and engineering know-how associated with leading-edge technology. To prolong the excess returns enabled by leading-edge process technology, or nodes, TSMC initially focuses on logic products, mostly used on central processing units and mobile chips, then focuses on more cost-conscious applications. This strategy has been successful, illustrated by the fact that the firm is one of the two foundries still possessing leading-edge nodes while dozens of peers lag.

We note two long-term growth factors for TSMC. First, the consolidation of semiconductor firms is expected to create demand for integrated systems made with the most advanced nodes. Second, organic growth of artificial intelligence, Internet of Things, and high-performance computing applications may last for decades. AI and HPC play a central role in quickly processing human and machine inputs to solve complex problems like autonomous driving and language processing, which accentuates the need for more energy-efficient chips. Cheaper semiconductors have made integrating sensors, controllers, and motors to improve home, office, and factory efficiency possible.

Bulls Say Phelix Lee, Senior Equity Analyst, 15 Jan 2026

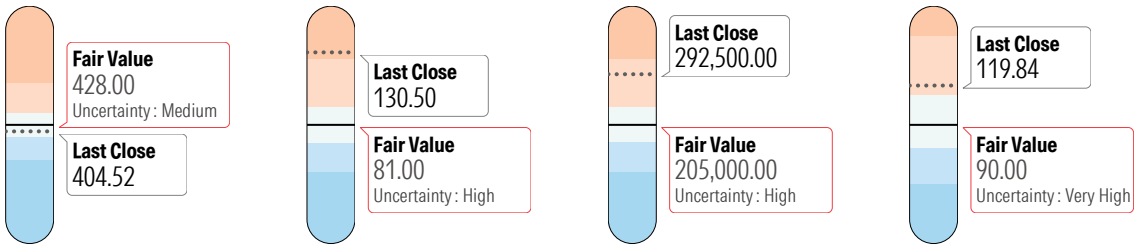
- TSMC should consistently earn higher gross margins than competitors thanks to its economies of scale and premium pricing justified by cutting-edge process technologies.
- TSMC wins when customers compete to offer the most advanced processing systems using the latest process technologies.
- TSMC will benefit from more semiconductor firms embracing the fabless business model and internet giants designing their own data center chips.

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Competitors



	Taiwan Semicon...ing Co Ltd ADR TSM	United Microelectronics Corp 2303	Samsung Electronics Co Ltd 005930	Intel Corp INTC
Economic Moat	Wide	None	None	None
Currency	USD	TWD	KRW	USD
Fair Value	428.00 22 May 2026 16:47, UTC	81.00 25 May 2026 08:25, UTC	205,000.00 22 May 2026 07:24, UTC	90.00 22 May 2026 16:43, UTC
1-Star Price	577.80	125.55	317,750.00	157.50
5-Star Price	299.60	48.60	123,000.00	45.00
Assessment	Fairly Valued 25 May 2026	Overvalued 26 May 2026	Overvalued 26 May 2026	Overvalued 25 May 2026
Morningstar Rating	★★★22 May 2026 05:00, UTC	★★25 May 2026 05:00, UTC	★★22 May 2026 05:00, UTC	★★22 May 2026 05:00, UTC
Analyst	Phelix Lee, Senior Equity Analyst	Phelix Lee, Senior Equity Analyst	Jing Jie Yu, Equity Analyst	Brian Colello, Senior Equity Analyst
Capital Allocation	Standard	Standard	Standard	Standard
Price/Fair Value	0.95	1.54	1.43	1.33
Price/Sales	15.78	6.41	4.42	10.51
Price/Book	11.42	3.89	4.08	5.40
Price/Earning	33.93	31.73	23.45	—
Dividend Yield	0.82%	2.28%	0.57%	0.00%
Market Cap	2,098.03 Bil	1,637.31 Bil	1,920,719.71 Bil	602.32 Bil
52-Week Range	190.03—421.97	40.10—133.50	53,700.00—300,500.00	18.97—132.75
Investment Style	Large Growth	Large Blend	Large Blend	Large Growth

Bears Say Phelix Lee, Senior Equity Analyst, 15 Jan 2026

- Although TSMC is the foundry leader, each generation of process technology matures and commoditizes quickly, forcing the company to deal with pricing pressure.
- TSMC's new approach to diversify production geographically may add cost pressures with little added resilience to stability.
- Samsung and Intel are committed to heavy capital spending under the support of the US government. SMIC and other state-supported Chinese foundries also lurk as potential threats.

Economic Moat Phelix Lee, Senior Equity Analyst, 16 Oct 2025

We believe TSMC's wide moat stems from intangible assets that are realized from its leading position in advanced process technologies, or nodes, in semiconductor manufacturing. It enjoys a cost advantage stemming from its ability to spread unit research and development costs, given its dominating scale.

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TSMC's longstanding leadership in node advancement comes from its ability to work with customers years in advance to identify the optimal improvements in power, performance, area, cost, and time to market, or PPACT, for each chip while maintaining fiscal discipline. This has justified higher prices than peers. TSMC's leadership creates a virtuous cycle of premium products, premium pricing, and profits for next-generation research and development. Through seizing opportunities in smartphones in the 2010s and data centers (including AI) in the 2020s, TSMC has increased its market share to about 70% of the foundry (contract manufacturing) market and simultaneously expanded operating margins to above 40%. We project TSMC to achieve low 20s returns on invested capital in the next five years, above its weighted average cost of capital of 8.2%.

We believe the gap between TSMC and its smaller peers remains wide. TSMC and Samsung Electronics are the only two foundries that sell 5- and 3-nanometer chips at scale to external customers, and Samsung has lost market share due to yield issues at 3 nm. Although Intel's 14A node may emerge as a viable competitor to TSMC's A14 node in 2028 with strong US government support, we think this will end up with Intel replacing Samsung as TSMC's alternative instead of dethroning TSMC. The gap is starker with smaller foundries, as the latter tend to stop at 14 nm or 28 nm nodes, which were first mass produced over a decade ago.

Another of TSMC's intangible assets is its close relationships with industry giants, which help justify investments in the next nodes. These relationships and TSMC's lead in advanced nodes convince customers to share detailed roadmaps, which reduces the odds of over/underinvestment and helps optimize efforts to improve chip PPACT for various applications like consumer electronics, high-performance computing, and automotives. TSMC is the only leading-edge foundry that does not have its own branded products and has limited licensed intellectual properties, so customers can take comfort in the lack of conflict of interest and equal access to new technology that TSMC offers.

TSMC enjoys structural cost advantages compared with Intel and Samsung, especially in R&D. TSMC only serves external customers, while Intel and Samsung have to firewall client-owned designs from in-house ones. The firewall comes with additional compliance costs and potential duplicate efforts in R&D (at least two sets of designs for each product). The second cost advantage is TSMC's massive scale (8 times Samsung, 6 times Intel Foundry) to push down burgeoning unit R&D costs for cutting-edge chips. The actual scale difference may be higher as we estimate TSMC has an overwhelming 90% share in AI computing chips, which makes catching up more difficult than it seems. TSMC also spends almost double on R&D versus Intel, which gives it a better chance to address diverse client needs in cutting-edge AI, smartphone, and autonomous driving. The third advantage is lower R&D personnel costs in Taiwan. It would be hard for Intel to match TSMC's costs, as it may face pushback from the government and other stakeholders if it outsources R&D to low-cost regions, including Taiwan.

TSMC shows signs of network effects and switching costs, but we reckon they are not durable enough

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to qualify as a moat source. Rather, we view them as customer considerations that reinforce TSMC's intangible assets. Surging costs of designing cutting-edge chips make it costly to maintain multisourcing arrangements. IBS estimates chip design costs to exceed USD 500 million from 3 nm onward, and historically, newer nodes have always cost more despite more automation. This means that when TSMC wins orders for a chip model, customers are disincentivized to switch suppliers in the middle of a product cycle, which could be one year for smartphone chipsets and two years for AI data centers. Automotive and industrial chips have longer product lifecycles, with some exceeding 10 years to qualify as a stand-alone switching costs moat source, but they are cheaper to design with the use of mature nodes, so multisourcing is common.

A side effect of TSMC's dominant market share is that the company doubles as a marketplace between electronic design automation firms and chip designers, which gives rise to some aspects of network effect. EDA firms offer intellectual property blocks that serve as ready-to-use portions of chip designs for mundane tasks and are specific to a single node of a single foundry. As a part of EDA firms' revenue is based on licensed fees per chip sold, they prioritize IP blocks that are likely to be widely adopted. The wide clientele of TSMC makes it attractive for EDA firms to prioritize compatibility with TSMC's processes, and TSMC's current and prospective customers will find the wide range of available IP blocks valuable for product flexibility. The marketplace is especially strong at AI data centers, where TSMC has around 90% market share and bundles in-house advanced packaging offerings.

Fair Value and Profit Drivers Phelix Lee, Senior Equity Analyst, 15 Jan 2026

Our base-case fair value estimate is USD 428 per ADR, at which TSMC would trade at a forward price/earnings ratio of 26 times per 2026 estimates. We use a weighted average cost of capital of 8.2% to discount our forecast cash flow for TSMC.

We project the company's top-line compound annual growth rate at 20.9% over the next five years. Even with its dominant market share, we believe TSMC can deliver above-industry growth through a higher proportion of more valuable 10 nm to 1.4 nm logic and 28 nm to 7 nm specialty products, which are currently only produced by it and Samsung at scale. We expect Internet of Things and automotive applications to be sources of incremental demand in newer specialty products. In terms of node advancement, mass production of A16 manufacturing is expected to begin in second half of 2026 and A14 in 2028.

We project 2026 gross and operating margins to be higher year on year at 61% and 50%, respectively, as profitability of 3 nm production improves and revenue of high-gross-margin AI products grows faster than expected. While quarterly margins may fluctuate while the company ramps up production of a new node, long-run margins should be stable, as we expect TSMC's moat to support its pricing power for years to come. Management also aligns its interests to the 56% gross margin target by listing it as

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one of the criteria for performance-linked bonuses.

High-performance computing is TSMC's largest growth driver. We believe increasing in-house design of cloud computing and AI chips by US and Chinese internet giants will benefit TSMC for the next few years.

One recurring threat to TSMC's long-term growth is the loss of key personnel to competitors. In the past, TSMC has managed the issue by offering above-average salaries to employees. But to counter Chinese semiconductor companies that are poaching talent, TSMC rolled out a performance share scheme in 2021, which links financial performance and environmental, social, and governance initiatives to staff remuneration. We think this better aligns employees' interests with more stakeholders. TSMC's scheme is one of Taiwan's first to include ESG goals.

Risk and Uncertainty Phelix Lee, Senior Equity Analyst, 16 Oct 2025

We assign TSMC a Medium Morningstar Uncertainty Rating. The company operates in the semiconductor industry, which is one of the most cyclical. TSMC derives about a third of its revenue from the smartphone market. The industry alternates between shortages and oversupply. Foundries cannot always raise prices during shortages but have to deal with high fixed costs in all downturns. Compared with its peers, TSMC's earnings volatility has been lower, with no earnings per share decline larger than 20% in the past 10 years. We expect this to continue as a result of TSMC's dominant share in high-end products and customers' preference for TSMC as their primary (sometimes sole) foundry.

TSMC has client concentration risk, with the largest customer contributing 22% of revenue in 2024 and the top four clients about 50%. We believe Apple has been TSMC's largest customer for the last five years, owing to consistent wins of A series processors on multiple devices. Due to short product lifecycles, the possibility of Apple choosing Samsung as the foundry for an upcoming chip lingers. The advent of AI may propel Nvidia to replace Apple as the larger customer in the next two or three years.

Currency risk is limited, as most transactions are made in US dollars. Intellectual property theft is a major risk. The most high-profile incident was TSMC's settlement with SMIC, in which the firm received shares and cash from SMIC after a series of legal disputes from 2003 to 2009, as reported by Reuters.

TSMC's expansion requires a lot of land, electricity, and water. Land acquisition may be slowed by objections from locals. The firm works with government agencies to ensure the supply of electricity and water, and with suppliers to enhance its waste and water treatment systems.

CEO C.C. Wei has doubled as chair since June 2024. While near-term implications should be minimal, we hope he can balance managing day-to-day operations with government relations.

Capital Allocation Phelix Lee, Senior Equity Analyst, 15 Jan 2026

We assign TSMC a Standard Morningstar Capital Allocation Rating, We view the company's decision in

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March to invest another USD 100 billion in Arizona production sites as not commercially driven. The decision contrasts with TSMC previously saying both construction and manufacturing are more expensive in the US than in Taiwan.

That said, there is no immediate danger to the company's 25% return on equity target as high-performance computing and autonomous driving remain strong growth drivers. TSMC's ROIC and ROE are far higher than those of UMC and SMIC, with the latter averaging less than 10% over the last 10 years. TSMC's earnings are also more stable than peers', with a 2012-25 EPS compound annual growth rate at 16% without major decreases (more than 20%) year on year. UMC had four major decreases and SMIC had three in the same period. We believe such impressive financial performance is evidence of management's ability to expand capacity without being distracted by short-term supply/demand imbalances and focus on cementing long-term technological leadership instead of pursuing short-term opportunistic pricing during shortages. Moreover, TSMC is more disciplined in expansion than peers. It tends to direct 30%-50% of revenue to capital expenditures each year. UMC and SMIC's capital expenditures appear to be more arbitrary, with historical capital expenditures/sales ratios fluctuating between 9% and 62% and between 31% and 150%, respectively. TSMC also appears to match capital expenditures with future demand better than its peers, with a more stable depreciation/sales ratio in the mid-20s.

TSMC's more stable earnings lead to more consistent dividends. The company has been paying dividends nonstop since 2004 (for 2003 earnings) without any cuts. In fact, TSMC's annual dividend per share has increased for six consecutive years with a payout ratio at around 50%, which is a feat, given the industry's cyclicality and heavy investments needed. The company has two main shareholder return policies. The first is to at least maintain, if not increase, the dividend per share every year. The second is to prioritize dividends over share repurchases. Dividends have been paid quarterly since 2019. We forecast dividends to increase to TWD 34 per share by 2030. In contrast, SMIC has never declared a dividend since its IPO in 2004, and UMC's dividend history is marked with ups and downs in line with its earnings.

TSMC has not made material acquisitions in recent years. This is reasonable, considering that other foundries look unattractive in the face of TSMC's unparalleled manufacturing capabilities, and the company has no plans to compete with customers in IC design or replace downstream packaging, assembly, and testing firms.

Analyst Notes Archive

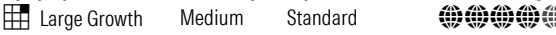
TSMC: Annual Symposium Sheds Light on Three-Year Road Map and Affirms Our Long-Term Confidence Phelix Lee,Senior Equity Analyst,24 Apr 2026

Taiwan Semiconductor Manufacturing Company held the first session of its annual symposium in

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Santa Clara, California. It unveiled a three-year road map on cutting-edge process nodes, including A13, A12, and N2U, and updates on market size, advanced packaging, and mature nodes. Why it matters: TSMC now expects the semiconductor market to reach USD 1.5 trillion by 2030 versus USD 1 trillion in 2025. The main change is an 83% increase in high-performance computing (like artificial intelligence) to USD 825 billion, implying over USD 1 trillion annual hyperscaler spending. TSMC confirmed that all announced nodes do not use high-numerical aperture extreme ultraviolet lithography tools. A13 will be backward compatible with A14, and N2U partly with N2P and N2. These two factors enable TSMC to offer chip improvements more cheaply and save customers' design costs. Advanced packaging upgrades center on stitching more chips together for computing power and introducing optical technology to reduce latency. The maximum stitch area could increase seven times by 2029, and data transmission can be up to 200 times faster. The bottom line: We retain our fair value estimates of TWD 2,700 (USD 428 per ADR) for wide-moat TSMC, as the symposium boosts our confidence in TSMC's ability to provide performance upgrades at a reasonable cost. The stock is attractive as the market is too worried about competition from Intel. We like how TSMC is outlining a two-pronged strategy to chip upgrades. The N2 to N2P to N2U path is a lower-cost route fit for mobile and autos. The A16 to A14 to A13 path is optimized for data throughput, performance, and high-power delivery. Between the lines: We believe TSMC is keeping its options open to adopt high-NA tools in the A10 node by being tight-lipped on A10. TSMC may order the EXE:5200D 2029 model from ASML to evaluate cost-effectiveness, just as it may have done for the first-generation high-NA tool.

TSMC Earnings: Refining Guidance and Expansion Plans Amid Strong AI Demand Phelix Lee,Senior Equity Analyst,16 Apr 2026

Taiwan Semiconductor Manufacturing Co. refined its 2026 US dollar revenue and capital expenditure guidance to the upper bounds of prior ranges. First-quarter revenue topped TWD 1.1 trillion (USD 35.9 billion), up 6% sequentially. Gross margin grew 746 basis points from the prior quarter to 62.3%.Why it matters: Capex guidance is now approaching the previous upper bound of USD 56 billion, while revenue growth guidance has been raised to above 30% from about 30%. Both support the strength of the near-term and three- to five-year demand outlook for artificial intelligence applications. We think the second wave of 3-nanometer capacity expansions in Taiwan, Arizona, and Japan, and management's view that 3 nm profitability will converge to the corporate average in the second half, imply manufacturing yields are mature enough to meet the requirements of large-size AI chips. Surging commodities and raw materials costs caused by the Iran war are factored into TSMC's outlook. We believe the financial impact from higher costs is secondary to the ramp-up of 2 nm chips and non-Taiwan production. We assume no material supply chain disruptions in our forecasts.The bottom line: We retain our TWD 2,700/USD 428 fair value estimate for wide-moat TSMC, as the earnings bring little change to our long-term view. The stock is attractive as the market is overly wary of Intel leveraging its packaging solutions to entice customers to switch. Investors expect Intel to win customers more easily through its EMIB

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Taiwan Semiconductor Manufacturing Co Ltd ADR TSM ★★★ 22 May 2026 21:47, UTC

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packaging offerings, threatening TSMC's long-term market share. We view these concerns as overblown, as TSMC is aggressively expanding advanced packaging capacity and developing new versions that improve system performance.Between the lines: The US' recently proposed bipartisan Match Act seeks to tighten equipment exports to China. We think it makes procurement for Japan's and Germany's specialty fabs slightly easier, as TSMC may not need to compete with aggressive Chinese buyers.

War in Middle East Raises Near-Term Risks for Semiconductor Players, but Long-Term Drivers Intact

Phelix Lee,Senior Equity Analyst,10 Mar 2026

Shares of Taiwan Semiconductor Manufacturing Company, Samsung Electronics, and SK Hynix have fallen by 9%-22% since the war in the Middle East broke out, as the escalating conflict threatens the supply stability of key energy sources such as crude oil and liquefied natural gas. Why it matters: We think rising energy prices, if prolonged, could be a near-term risk for chip manufacturers. Higher energy costs for AI data centers could slow AI infrastructure buildouts, while fabs in Taiwan and South Korea would face growing cost pressures from higher LNG prices. Crude oil makes up 38% of the energy needs for the US, home to most global AI data centers. AI data centers are three to five times more power-hungry than regular counterparts, so prolonged crude price hikes could slow AI server buildouts and consequently AI chip demand, as the total cost of ownership could rocket. We estimate energy costs for TSMC, Samsung, and SK Hynix made up 3%-6% of 2025 revenue. Should the war be prolonged, we could see these costs rise materially. However, we think these costs can be passed on to customers, given the broad undersupply of AI-related chips. The bottom line: We maintain our fair value estimates for wide-moat TSMC at TWD 2,700 and USD 428, at KRW 140,000 and USD 2,443 for no-moat Samsung Electronics, and at KRW 623,000 for no-moat SK Hynix. We will review our estimates if the war shows signs of being prolonged. Key stats: Korea and Taiwan import about 15% and 33% of LNG from Qatar and maintain at least nine and 11 days in reserves, respectively. Without mitigating efforts, depletion risks would be five to eight weeks away. LNG is used for almost 25% and 50% of Korea and Taiwan's electricity generation, respectively. While Taiwan imports over 70% of oil from the Middle East, the risk is less acute as Taiwan maintains about four months of reserves, and Kuo Jyh-Huei, former minister of economic affairs, said in 2025 that only 20% of imports go through the Strait of Hormuz.

TSMC Earnings: Lifting Fair Value Estimate as Buoyant Guidance Backed by Voracious AI Demand

Phelix Lee,Senior Equity Analyst,15 Jan 2026

Taiwan Semiconductor Manufacturing guided its 2026 USD revenue to grow by 30% and capital expenditure as high as USD 56 billion. Its December quarter revenue topped TWD 1 trillion (USD 33.7 billion), up 6% sequentially. Gross margin grew 287 basis points from the prior quarter to 62.3%. Why it matters: The 2026 capex budget of USD 52 billion-USD 56 billion crushed our prior expectation of USD 47 billion. Management attributes this to exuberant artificial intelligence demand and backed this view

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with a 2024-29 revenue CAGR of mid-20s (previously 20%) and an AI revenue CAGR of over 50% (mid-40s). Management's track record of conservative guidance, plus its thinly veiled reference to the benefits of an indirect customer (Meta) firm boost our confidence that TSMC can incur TWD 8.8 trillion in sales by 2029, or 23.3% 2025-29 CAGR. We were previously wary of the ripple effects that OpenAI's cash burn posed risks to TSMC's revenue targets. Now we have more visibility on AI spending since Google is increasing availability of tensor processing units and Broadcom is adding customers. The bottom line: We increase our fair value estimate for TSMC to TWD 2,700 from TWD 1,900 (USD 428 per ADR from USD 310) on higher expectations of durable AI spending. We believe the market is skeptical on how specialized chips can complement more versatile Nvidia chips. We are more bullish in China AI demand as Nvidia has secured export licenses in the US, and Chinese AI firms thirst for the most power-efficient chips made on TSMC processes. We lifted our 2026-29 revenue and EPS estimates by 10%-45% and 7%-53%, respectively. We believe TSMC is the main beneficiary among all chipmakers, given its proven technological leadership and reliability. Big picture: TSMC's increased capex is music to the ears of wafer fab equipment makers like Lam Research, ASML, and Tokyo Electron. We believe WFE firms are fit for hedging against competition from Intel Foundry and Taiwan-specific political risks.

TSMC Earnings: Lifting Valuation by 6% as Insatiable AI Demand Comes From Multiple Customers

Phelix Lee,Senior Equity Analyst,16 Oct 2025

TSMC raised its full-year US dollar revenue growth outlook to the mid-30s from 30% and the lower bound of capital expenditure by 5% to USD 40 billion. Third-quarter revenue was TWD 990 billion (USD 33.1 billion), up 6% sequentially. Gross margin grew 84 basis points from the prior quarter to 59.5%.Why it matters: Strong investments in artificial intelligence and a rebound in mature process nodes support TSMC's higher full-year outlook. We expect cloud service providers and other customers to ramp up AI spending, which eases TSMC's exposure to Nvidia. Given OpenAI's multiple agreements with TSMC's customers, we model 2026 capex to be USD 47 billion, up 12% from 2025. We do not expect the full plan of OpenAI to be realized, as we are wary of that company's cash burn, and there is no way to learn how OpenAI will pay for the chips. We only make negligible changes to gross margin forecasts despite higher revenue, as we believe high capex is needed in both Arizona and Taiwan to fully capture AI demand. TSMC will have to spend extra to accelerate 2-nanometer capacity in Arizona.The bottom line: We hike our fair value estimate for TSMC to TWD 1,900 per share from TWD 1,800 (and to USD 310 per ADR from USD 306) on better guidance. The shares appear cheap, as we expect demand from AI inference will be long-lasting, potentially more than training AI modelsLong view: Management prefers to update AI growth in its January call and reiterated a mid-40s compound annual growth rate from 2024 to 2029. We estimate the 2025-29 CAGR for AI revenue to be about 34% and expect AI will contribute 46% of revenue by 2029.Bears say: OpenAI can be a sectorwide risk as it relies on external funds to run. Reuters reported that OpenAI is on track for USD 13 billion in revenue for 2025, but the annual contract value of announced deals is tens of times larger.

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TSMC: Nvidia's Collaboration With Intel Has Minimal Impact on Long-Term Outlook; Shares Undervalued Phelix Lee,Senior Equity Analyst,19 Sep 2025

The share price of Taiwan Semiconductor Manufacturing Company has been steady amid Intel and Nvidia's plan to codevelop multiple generations of custom data center and PC products. Nvidia will acquire USD 5 billion of Intel's common stock at USD 23.28 per share. Why it matters: Nvidia has not committed to shifting massive portions of its chip production to Intel Foundry from TSMC. Given the lack of a timeframe for Intel and Nvidia's plan, we believe there's minimal impact on TSMC's 2-nanometer and A16 outlook. Although Intel has raised USD 15.9 billion since Lip-Bu Tan became CEO in March, TSMC still has larger scale to keep unit development costs low. The funds raised pale in comparison to TSMC's USD 40 billion annual capital expenditure budget. The bottom line: We maintain our TWD 1,800 (USD 306 per ADR) fair value estimates for TSMC. The company remains our top pick given its quasi monopoly on advanced chips. We expect that a broader clientele in artificial intelligence will also limit the downside of Nvidia using Intel Foundry more. Between the lines: Nvidia's stake may be useful for Intel to complete intellectual property development with Synopsys. The IP blocks are useful for customers designing products for Intel's 14A node. However, Nvidia seems to be testing the waters at Intel as it retains the option to mass-produce next-generation products using TSMC's A14 node. Separately, Qualcomm's CEO Cristiano Amon mentioned that Intel has yet to roll out suitable nodes for them, which we see as reassurance of TSMC's outlook.

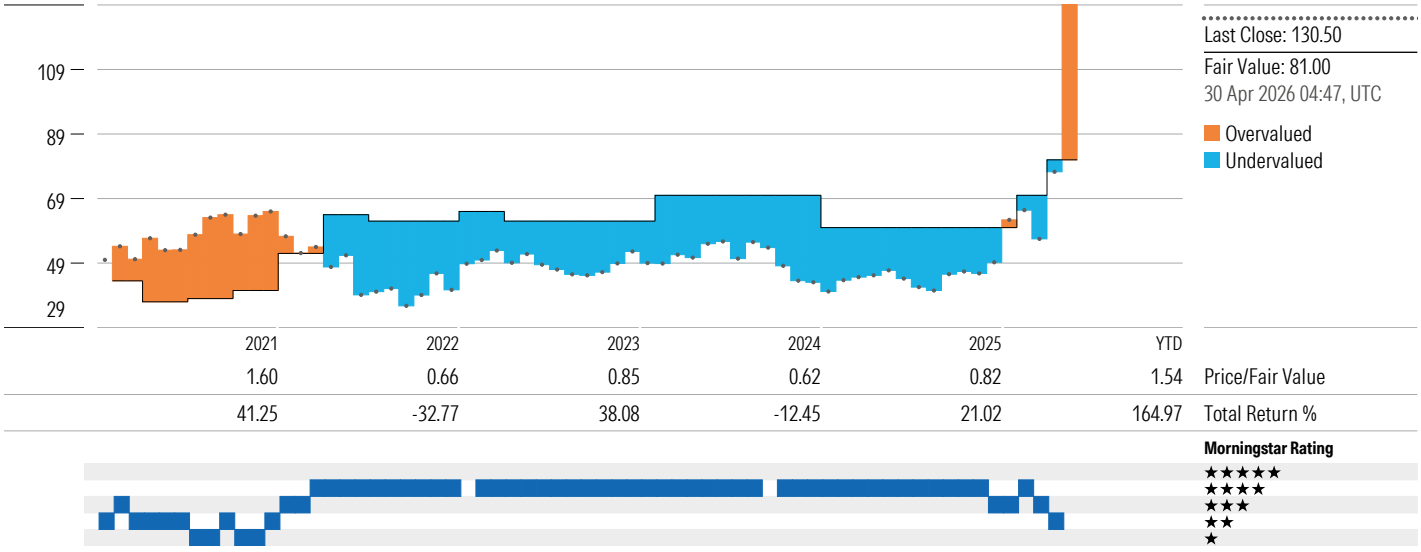
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Taiwan Semiconductor Manufacturing Co Ltd ADR TSM ★★★ 22 May 2026 21:47, UTC

Competitors Price vs. Fair Value

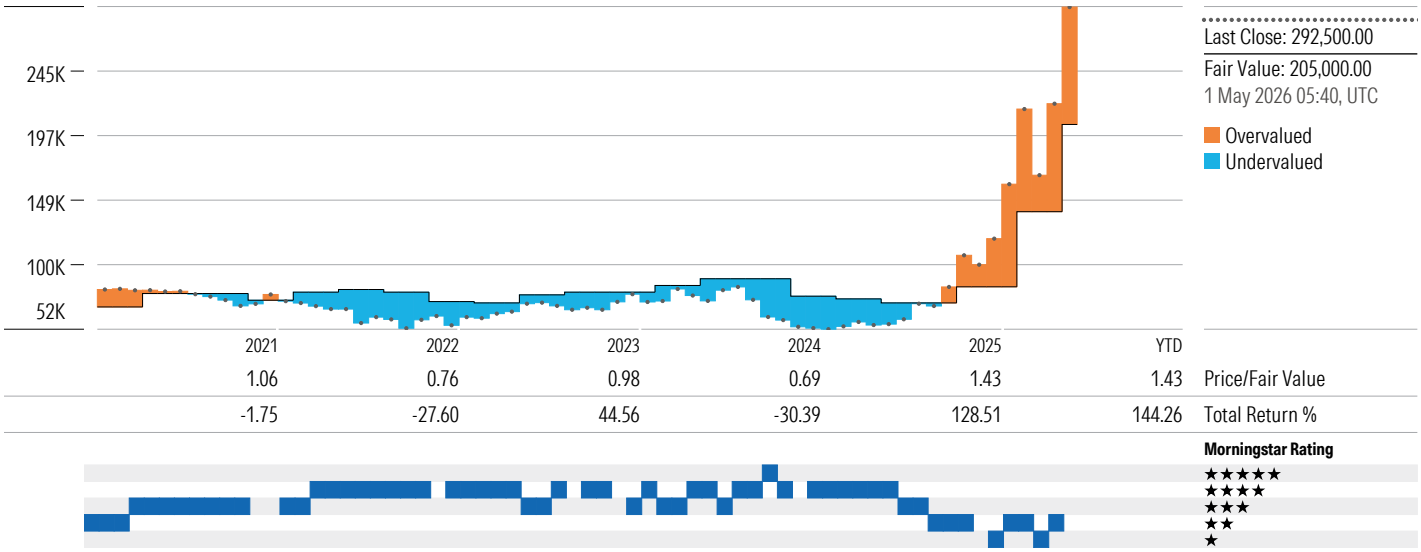
United Microelectronics Corp 2303



	2021	2022	2023	2024	2025	YTD	
	1.60	0.66	0.85	0.62	0.82	1.54	Price/Fair Value
	41.25	-32.77	38.08	-12.45	21.02	164.97	Total Return %

Total Return % as of 26 May 2026. Last Close as of 26 May 2026. Fair Value as of 30 Apr 2026 04:47, UTC.

Samsung Electronics Co Ltd 005930



	2021	2022	2023	2024	2025	YTD	
	1.06	0.76	0.98	0.69	1.43	1.43	Price/Fair Value
	-1.75	-27.60	44.56	-30.39	128.51	144.26	Total Return %

Total Return % as of 22 May 2026. Last Close as of 22 May 2026. Fair Value as of 1 May 2026 05:40, UTC.

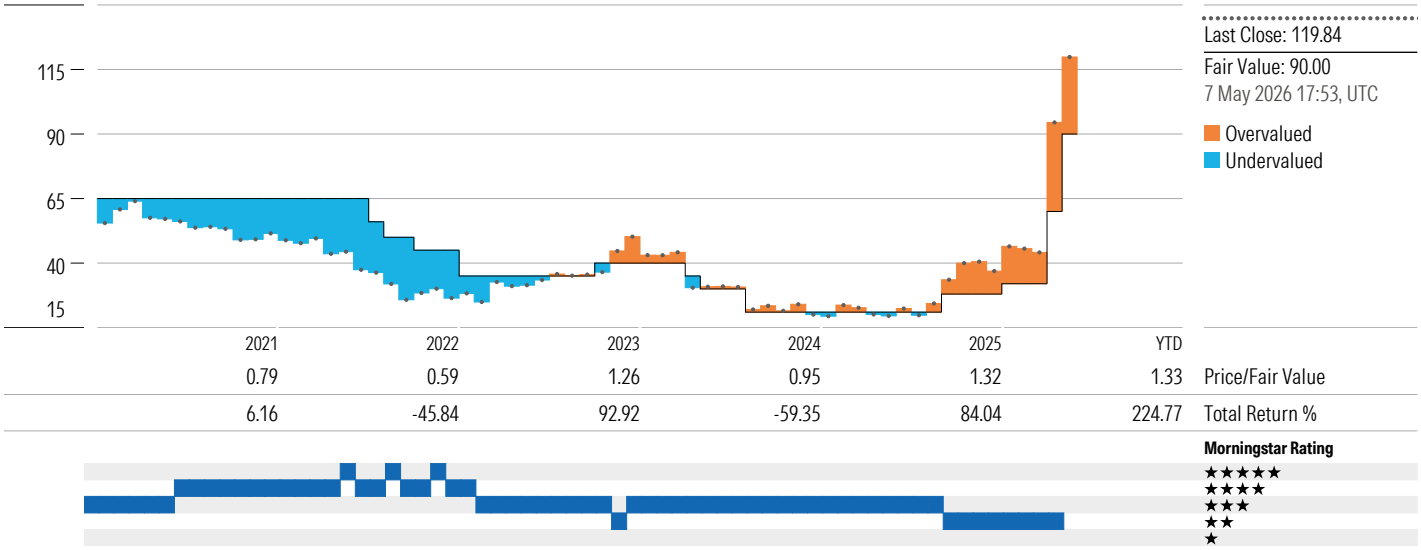
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Competitors Price vs. Fair Value

Intel Corp INTC



Total Return % as of 22 May 2026. Last Close as of 22 May 2026. Fair Value as of 7 May 2026 17:53, UTC.

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Morningstar Valuation Model Summary

Financials as of 15 Jan 2026	Actual			Forecast				
Fiscal Year, ends 31 Dec	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Revenue (TWD Bil)	2,162	2,894	3,809	4,879	6,164	7,421	8,813	9,839
Operating Income (TWD Bil)	921	1,322	1,827	2,443	3,013	3,669	4,337	4,744
EBITDA (TWD Bil)	1,463	1,991	2,525	3,331	4,078	4,946	5,874	6,567
Adjusted EBITDA (TWD Bil)	1,463	1,991	2,525	3,331	4,078	4,946	5,874	6,567
Net Income (TWD Bil)	815	1,264	1,696	2,136	2,727	3,329	3,932	4,324
Adjusted Net Income (TWD Bil)	815	1,264	1,696	2,136	2,727	3,329	3,932	4,324
Free Cash Flow To The Firm (TWD Bil)	26	209	406	806	682	784	851	898
Weighted Average Diluted Shares Outstanding (Bil)	26	26	26	26	26	26	26	26
Earnings Per Share (Diluted) (TWD)	31.43	48.75	65.40	82.37	105.18	128.37	151.63	166.76
Adjusted Earnings Per Share (Diluted) (TWD)	31.43	48.75	65.40	82.37	105.18	128.37	151.63	166.76
Dividends Per Share (TWD)	11.00	11.25	17.00	23.00	26.00	30.00	32.00	34.00

Margins & Returns as of 15 Jan 2026		Actual			Forecast					
	3 Year Avg	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**5 Year Avg**
Operating Margin %	45.7	42.6	45.7	47.9	50.1	48.9	49.4	49.2	48.2	49.3
EBITDA Margin %	—	67.7	68.8	66.3	68.3	66.2	66.6	66.7	66.7	—
Adjusted EBITDA Margin %	—	67.7	68.8	66.3	68.3	66.2	66.6	66.7	66.7	66.9
Net Margin %	42.0	37.7	43.7	44.5	43.8	44.3	44.9	44.6	44.0	44.3
Adjusted Net Margin %	42.0	37.7	43.7	44.5	43.8	44.3	44.9	44.6	44.0	44.3
Free Cash Flow To The Firm Margin %	6.4	1.2	7.2	10.7	16.5	11.1	10.6	9.7	9.1	11.4

Growth & Ratios as of 15 Jan 2026		Actual			Forecast					
	3 Year CAGR	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**5 Year CAGR**
Revenue Growth %	18.9	-4.5	33.9	31.6	28.1	26.3	20.4	18.8	11.7	20.9
Operating Income Growth %	17.0	-19.2	43.5	38.2	33.7	23.3	21.8	18.2	9.4	21.0
EBITDA Growth %	18.3	-8.0	36.1	26.8	31.9	22.4	21.3	18.8	11.8	21.2
Adjusted EBITDA Growth %	16.7	-8.0	36.1	26.8	31.9	22.4	21.3	18.8	11.8	21.1
Earnings Per Share Growth %	16.0	-25.0	55.1	34.2	26.0	27.7	22.0	18.1	10.0	20.6
Adjusted Earnings Per Share Growth %	16.0	-25.0	55.1	34.2	26.0	27.7	22.0	18.1	10.0	20.6

Valuation as of 15 Jan 2026	Actual			Forecast				
	2023	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Price/Earning	20.3	26.6	29.2	30.9	24.2	19.8	16.8	15.3
Price/Sales	7.1	9.6	10.6	11.9	9.4	7.8	6.6	5.9
Price/Book	4.8	7.8	9.1	9.5	7.3	5.7	4.5	3.6
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	10.1	13.0	15.3	—	—	—	—	—
EV/EBIT	16.0	19.6	21.1	—	—	—	—	—
Dividend Yield %	1.7	0.9	0.9	0.9	1.0	1.2	1.3	1.3
Dividend Payout %	35.0	23.1	26.0	27.9	24.7	23.4	21.1	20.4
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 15 Jan 2026	Actual			Forecast				
Fiscal Year, ends 31 Dec	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
ROA %	14.7	18.9	21.4	22.5	23.6	23.5	22.8	20.8
ROE %	23.4	29.2	31.1	30.5	30.1	28.7	26.7	23.8
ROIC %	17.3	21.0	26.4	29.0	28.8	27.5	25.7	22.6

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Financial Leverage (Reporting Currency)	Actual			Forecast				
Fiscal Year, ends 31 Dec	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Debt/Capital %	5.9	3.6	—	1.3	1.2	1.1	1.1	1.0
Assets/Equity	1.6	1.5	1.5	1.4	1.3	1.2	1.2	1.1
Net Debt/EBITDA	-0.5	-0.7	—	-0.6	-0.7	-0.9	-1.0	-1.2
Total Debt/EBITDA	0.7	0.5	—	0.3	0.2	0.2	0.2	0.2
EBITDA/ Net Interest Expense	-137.1	-25.9	-27.0	-36.5	-53.0	-47.5	-48.8	-41.6

Forecast Revisions as of 15 Jan 2026	2026		2027		2028	
Prior data as of 16 Oct 2025	**Current**	**Prior**	**Current**	**Prior**	**Current**	**Prior**
Fair Value Estimate Change (Trading Currency)	428.00	1,897.03	—	—	—	—
Revenue (TWD Bil)	4,879	3,806	6,164	4,435	7,421	4,961
Operating Income (TWD Bil)	2,443	1,825	3,013	2,167	3,669	2,306
EBITDA (TWD Bil)	3,331	2,575	4,078	3,036	4,946	3,299
Net Income (TWD Bil)	2,136	1,607	2,727	1,992	3,329	2,108
Earnings Per Share (Diluted) (TWD)	82.37	61.97	105.18	76.82	128.37	81.28
Adjusted Earnings Per Share (Diluted) (TWD)	82.37	61.97	105.18	76.82	128.37	81.28
Dividends Per Share (TWD)	23.00	18.00	26.00	20.00	30.00	21.00

Key Valuation Drivers as of 15 Jan 2026

Cost of Equity %	7.5
Pre-Tax Cost of Debt %	5.5
Weighted Average Cost of Capital %	8.2
Long-Run Tax Rate %	17.0
Stage II EBI Growth Rate %	8.0
Stage II Investment Rate %	32.0
Perpetuity Year	20

Additional estimates and scenarios available for download at https://pitchbook.com/.

Discounted Cash Flow Valuation as of 15 Jan 2026

	TWD Mil
Present Value Stage I	3,175,181
Present Value Stage II	28,565,630
Present Value Stage III	36,527,417
Total Firm Value	**68,268,229**
Cash and Equivalents	2,470,759
Debt	998,415
Other Adjustments	130,000
Equity Value	**69,870,573**
Projected Diluted Shares	26
Fair Value per Share (USD)	**428.00**

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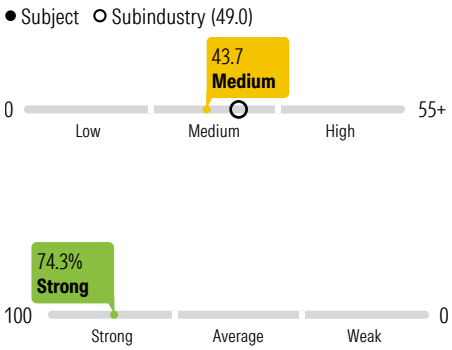
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Taiwan Semiconductor Manufacturing Co Ltd ADR TSM ★★★ 22 May 2026 21:47, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
404.52 USD 22 May 2026	428.00 USD 15 Jan 2026 19:57, UTC	0.95	2.10 USD Tril 26 May 2026	Wide	Large Growth	Medium	Standard	26 May 2026 05:00, UTC

ESG Risk Rating Breakdown

Exposure	
Company Exposure[1]	43.7
– Manageable Risk	40.6
Unmanageable Risk[2]	**3.1**
Management	
Manageable Risk	40.6
– Managed Risk[3]	30.1
Management Gap[4]	**10.4**
Overall Unmanaged Risk	**13.6**



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 74.3% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment[5]



ESG Risk Rating is of May 23, 2026. Highest Controversy Level is as of May 08, 2026. Sustainalytics Subindustry: Semiconductor Design and Manufacturing. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 26 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure			Management			ESG Risk Rating		
Taiwan Semiconduct...nufacturing Co Ltd	43.7 | Medium	0	55+	74.3 | Strong	100	0	13.6 | Low	0	40+
Intel Corp	50.7 | Medium	0	55+	68.0 | Strong	100	0	19.1 | Low	0	40+
Samsung Electronics Co Ltd	37.5 | Medium	0	55+	71.5 | Strong	100	0	12.6 | Low	0	40+
United Microelectronics Corp	47.7 | Medium	0	55+	68.3 | Strong	100	0	17.6 | Low	0	40+
GLOBALFOUNDRIES Inc	54.4 | Medium	0	55+	73.4 | Strong	100	0	17.7 | Low	0	40+

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Appendix

Historical Morningstar Rating

Taiwan Semiconductor Manufacturing Co Ltd ADR TSM 22 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
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Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
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United Microelectronics Corp 2303 25 May 2026 05:00, UTC

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Samsung Electronics Co Ltd 005930 22 May 2026 05:00, UTC

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Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
—	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★	★★

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Intel Corp INTC 22 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
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Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★

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Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, indepth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss shortterm market-price movements), but we believe these negatives are mitigated by deep analysis and our longterm approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

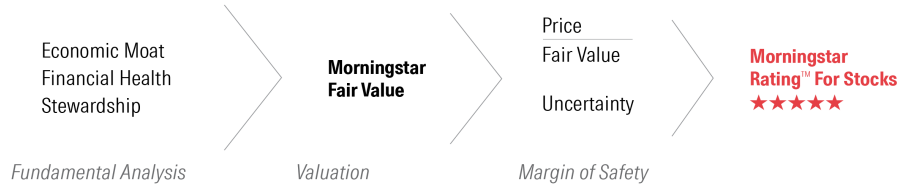
Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



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Research Methodology for Valuing Companies

outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

	Margin of Safety	
Qualitative Analysis Uncertainty Ratings	★★★★★Rating	★Rating
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

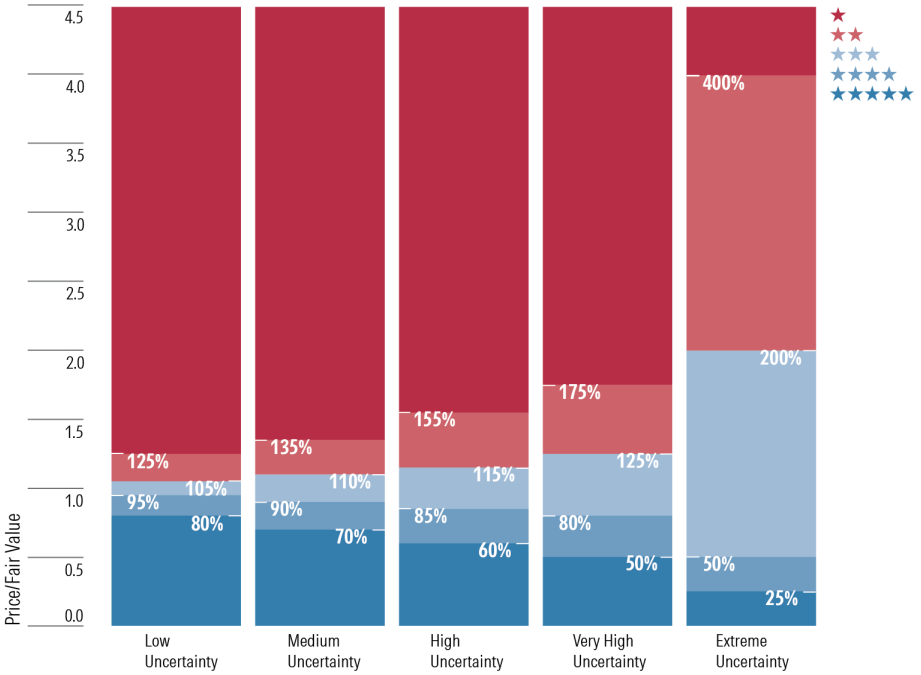
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to https://shareholders.morningstar.com

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multiyear time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

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